3 · Shaping the Lookback

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2 · Testing a Signal left two blanks inside its own definition. Writing momentum as $\mathrm{Avg}(r_{s,t-i})$, $i = 1 \ldots N$ fixes neither the length of the window nor the weight each return carries inside it—a flat average gives every return in the window an equal vote regardless of age, and commits to one window that cannot be both stable and timely.

An **exponentially weighted moving average (EWMA)** answers the first blank: replace the flat weights with a decay, so the newest return counts for the most and the average never quite forgets (§3.1). **Moving average convergence/divergence (MACD)** answers the second blank on top of that fix—instead of one EWMA it carries two, a fast one and a slow one, and trades the gap between them (§3.2, §3.3). MACD is therefore not a third technique standing beside EWMA; it is EWMA's own decay run twice, at two speeds, and read as a difference.

Both pieces run into the same wall no matter how they are tuned, and §3.4 is that wall: **a failure mode that survives every setting of either**, because it belongs to the market rather than to the parameters. The repair §3.4 can offer is local; the repair it cannot offer is 4 · Volatility Regimes.

3.1 EWMA instead of a simple moving average

3.1.1 Why the newest return should weigh most

A momentum signal is a weighted sum of past returns, and a simple moving average is one particular choice of those weights: make every one of them equal. Stated that way it is a claim about information—that a return from sixty days ago tells you exactly as much about where the asset stands today as yesterday's does.

It does not. **The newest return is the most recent evidence about the state the asset is in now**, so it should carry the most weight, and a return from three months ago the least. An exponentially weighted moving average is that preference written down.

Two things change at once. Relative to the box the newest lags are **over-weighted** and the old ones **under-weighted**—and the box's hard edge at twenty-one days is gone, so an EWMA thins out without ever quite forgetting.

3.1.2 The recursion, and the weights it produces

Definition 3.1 (EWMA)**.** Carry one running number and update it as each observation arrives:

$$\mathrm{EWMA}_t = (1-\lambda)\, a_t + \lambda\, \mathrm{EWMA}_{t-1}$$

where a_t is the newest observation and $\lambda \in (0,1)$ the decay—how much of the old average survives each step.

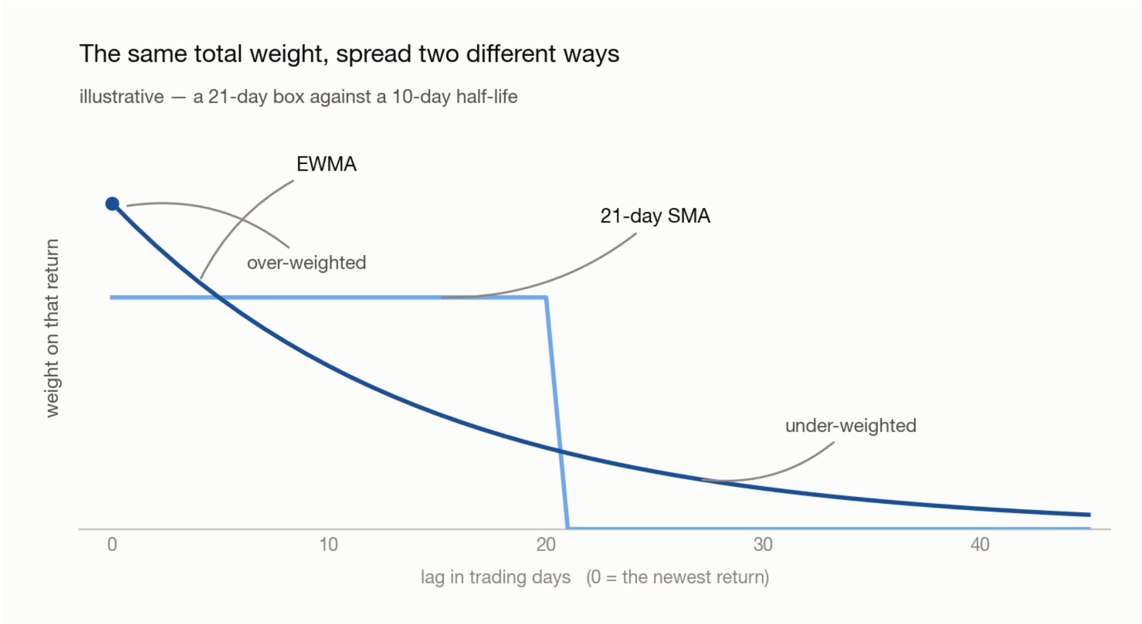

Figure 3.1: Weight carried by each past return against its lag in trading days, both curves normalised to the same total. The 21-day simple moving average is a flat box—identical weight out to twenty-one days, then nothing. The EWMA starts highest at lag zero, decays smoothly, and stays above zero past forty-five days. Illustrative.

Example 3.2. Take $\lambda = 1/2$ and three observations a_1, a_2, a_3, oldest first. Two updates run the whole thing:

$$\begin{aligned} \text{after } a_2: &\quad \frac{a_1 + a_2}{2} \\ \text{after } a_3: &\quad \frac{1}{2} \cdot \frac{a_1 + a_2}{2} + \frac{a_3}{2} = \frac{a_3}{2} + \frac{a_2}{4} + \frac{a_1}{4} \end{aligned}$$

Observation	**Age**	**Weight it ends up with**
a_3	newest	**1/2**
a_2	one step back	1/4
a_1	oldest	1/4

The newest observation carries half the total on its own, and each step back halves what is left—which is what *exponential* names. The oldest two tie only because the recursion had to start somewhere; run it long enough and that seam washes out.

3.1.3 Choosing how fast it decays

Tune the **half-life** H—the lag at which a return's weight has decayed to half the weight given to the newest one. Candidates are fractions of the lookback (1/2, 1/3, 1/5, 1/8), and—like every free parameter in this chapter—they are grid-searched rather than chosen; §3.2.2 grid-searches the analogous ratio for the fast/slow pairing §3.2 builds. In pandas the whole signal is `returns.ewm(halflife=H).mean()`.

No value of H is standard, and whatever a package ships as its default is an **empirical solution**—a number that fit somebody's historical data once. Every parameter in this chapter has that status, which is 5 · Modelling's subject rather than something to accept on authority.

3.2 Combining a fast and a slow horizon

§3.1 fixed the weight-shape blank with a decay. This section fixes the other one—not by finding a better single window, but by refusing to settle for just one.

3.2.1 Why two windows beat one

One lookback forces a choice between stable and timely. A long window rides a trend without being shaken out of it but arrives late at both ends; a short one turns on time, and turns on noise too. Rather than choose, carry both and trade **the gap between them**.

That gap is a single series—the fast average minus the slow one—and the rule is its sign: long while the fast average sits above the slow one, out when it drops back under. Two windows go in and one number comes out, which is what makes the pair an indicator rather than two rules run side by side.

What the crossing means. The fast average overtaking the slow one is not only an arithmetic consequence of the shorter window. It is the first evidence that flow has changed direction—that whoever was selling has stopped, or something large has started buying—while the move is still too small to register over a quarter. The same reading runs the other way and matters more there: the fast average rolling back under the slow one is what a large holder beginning to leave looks like from outside. Waiting for the slow window to turn on its own means selling to them on the way out.

3.2.2 Finding the pairing

The pairing is found by experiment, not chosen. Put the slow lookback N_s on one axis and the fast one N_f on the other, run 2's bar-plot test in every cell, and read the grid as a heat map.

Two things are true of that grid before any data is collected. **Half of it is empty by construction**—$N_f < N_s$ or there is no fast leg, so everything on and above the diagonal is struck out. And the candidates should be spaced **geometrically** rather than evenly: a day, a week, two weeks, a month, a quarter. Even spacing spends most of the grid on pairs that differ by a rounding error.

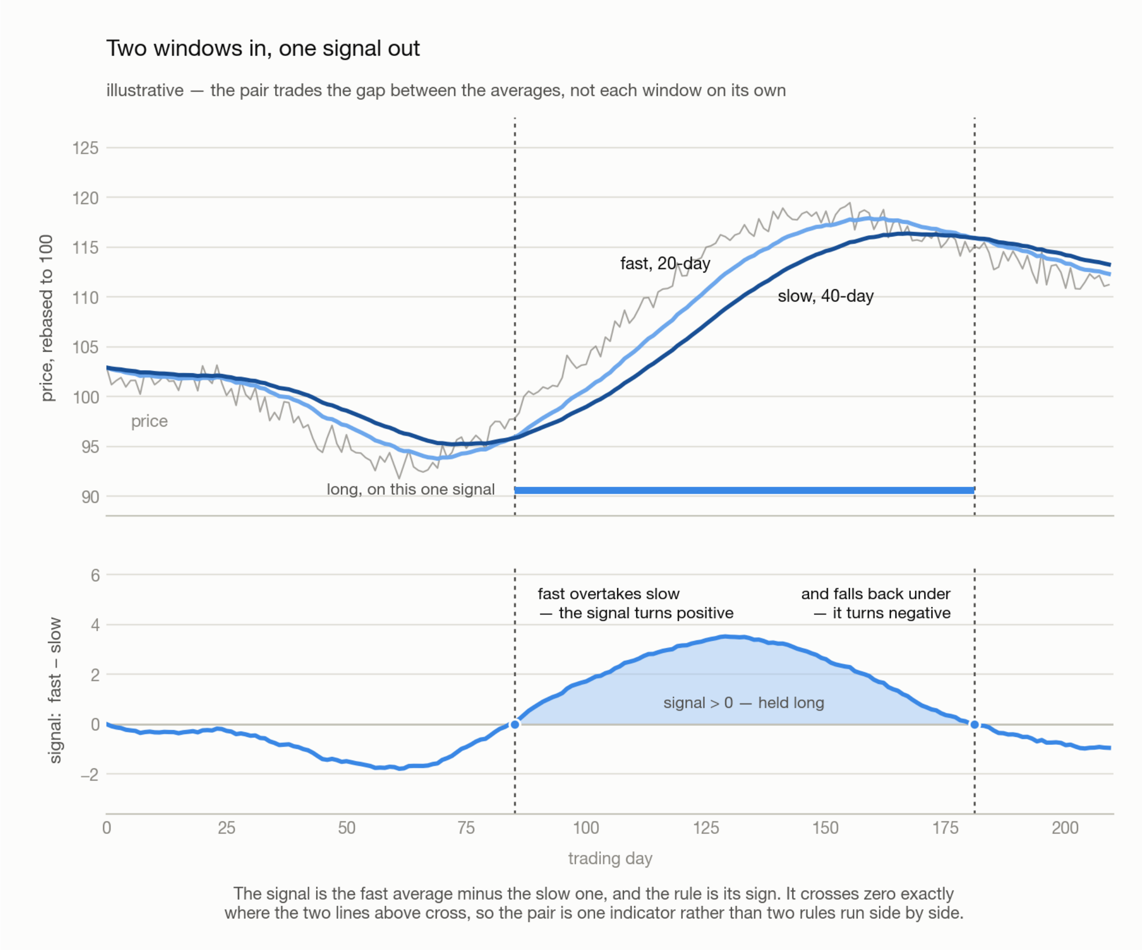

Figure 3.2: Above, one price path with a 20-day and a 40-day moving average over it; the fast average crosses above the slow one, and later drops back under, and the bar between those two dates marks the stretch the rule is long. Below, the single series the pair produces—fast minus slow—against zero, shaded where it is positive. Its two zero crossings fall exactly on the two dates the averages cross above. Illustrative.

What a diagonal band means. The warm cells run diagonally, and that shape is the finding rather than any single cell in it. On a geometric ladder a diagonal is a constant **ratio**, so what carries is N_s/N_f and not either window on its own—which is why the answer is quoted as a ratio at all. Research lands it near **2:1**, and MACD's 26/12 is one instance; for daily-to-weekly holding the band shows up around two weeks against a month. An empirical regularity, not a law, and 5 · Modelling's subject the moment you start trusting it.

Note (One hot cell is not a band)**.** A single dark square with pale neighbours is an artifact of however many parameter combinations were tried, not an edge. What earns belief is a contiguous warm region, because a real effect cannot switch off between one week and eight days.

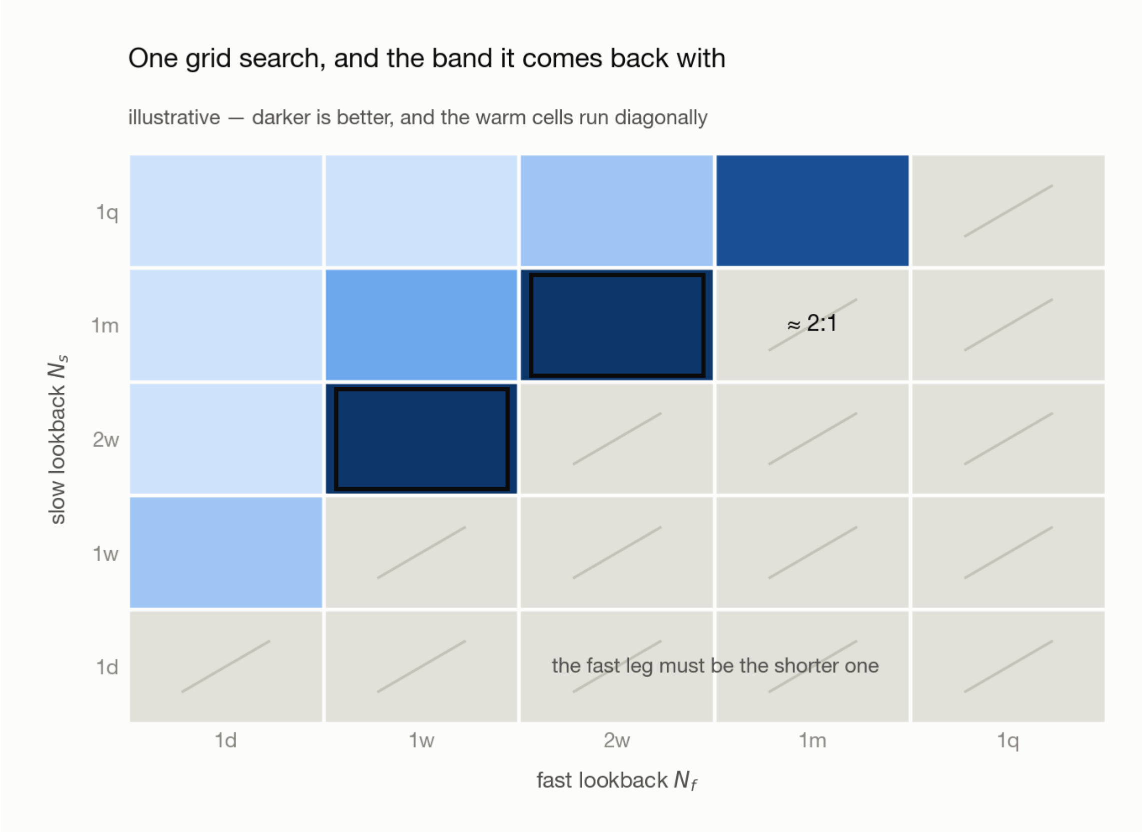

Figure 3.3: A five-by-five grid with a geometric ladder of lookbacks—one day, one week, two weeks, one month, one quarter—on both axes: fast across, slow up. Every cell on and above the diagonal is struck out, the fast leg having to be the shorter one. Of the rest, the darkest run in a diagonal band, and the two annotated cells sit at roughly two to one. Illustrative.

3.2.3 Where the combination pays

Best in **commodities**, and the reason is what moves them. An equity price answers to earnings, to the firm's own performance, to whoever is running it—many small idiosyncratic forces, none of which trend for long. A commodity answers to global energy supply, shipping, and whether the large participants are in the market at all, and it is far too large for one fund to push on its own, so a move that starts tends to be everyone moving together. Equities second. **Bonds** weakest: lean on one hard enough and someone takes the other side and presses it back.

3.3 MACD, stated precisely

§3.1 gave a window a decay; §3.2 took the difference of a fast average and a slow one and read its sign. That difference already has a name, and two further series conventionally travel with it.

3.3.1 The three series

Written out, MACD is three series built from two exponential moving averages of the **price**.

Definition 3.3 (MACD)**.** For a fast and a slow span $n_f < n_s$, each with its own smoothing constant $\alpha = 2/(\text{span}+1)$:

$$\begin{aligned} \text{MACD}_t &= \text{EMA}_{n_f}(P)_t - \text{EMA}_{n_s}(P)_t \\ \text{signal}_t &= \text{EMA}_9(\text{MACD})_t \\ \text{hist}_t &= \text{MACD}_t - \text{signal}_t \end{aligned}$$

where P_t is the price on date t and $\text{EMA}_n(P)_t$ its exponential moving average over span n. The asset subscript is dropped throughout—MACD is computed one asset at a time. Conventionally $n_f = 12$, $n_s = 26$, and 9 for the signal line.

Series	Reads as	Sign means
MACD line	recent average price versus a longer one	trend direction
Signal line	a smoothed MACD	the level MACD is crossing
Histogram	MACD minus its own smoothing	trend acceleration

3.3.2 Why it is momentum, not a separate indicator

Claim 3.4. MACD is a momentum signal. It is a weighted sum of past returns, differing from a lookback mean only in the shape of the weights.

Proof. Each EMA is a weighted average of past prices whose weights sum to one, so their difference weights the price i days ago by c_i, and those weights sum to **zero**:

$$\text{MACD}_t = \sum_i c_i P_{t-i}, \qquad c_i = \alpha_f(1-\alpha_f)^i - \alpha_s(1-\alpha_s)^i, \qquad \sum_i c_i = 0.$$

A zero-sum weighting is unchanged when every price shifts by a constant, so subtract P_t from each term and write $P_t - P_{t-i}$ as a sum of one-period changes $\Delta_{t-j} = P_{t-j} - P_{t-j-1}$. Collecting the coefficient of the change at lag j gives the **kernel** k_j:

$$\text{MACD}_t = \sum_j k_j \Delta_{t-j}, \qquad k_j = \sum_{i \le j} c_i = (1-\alpha_s)^{j+1} - (1-\alpha_f)^{j+1}.$$

Since $\alpha_f > \alpha_s$, every $k_j \ge 0$: MACD is a non-negative weighted sum of past price changes, exactly like a lookback mean. □

Note (What actually differs)**.** The kernel. A 21-day momentum weights the last 21 returns equally and everything older at zero; MACD's weights rise from a small value at lag 0, peak around lag 8, and decay without ever reaching zero. It therefore discounts *yesterday* relative to last week—deliberately, since the newest return is the noisiest—and never fully forgets.

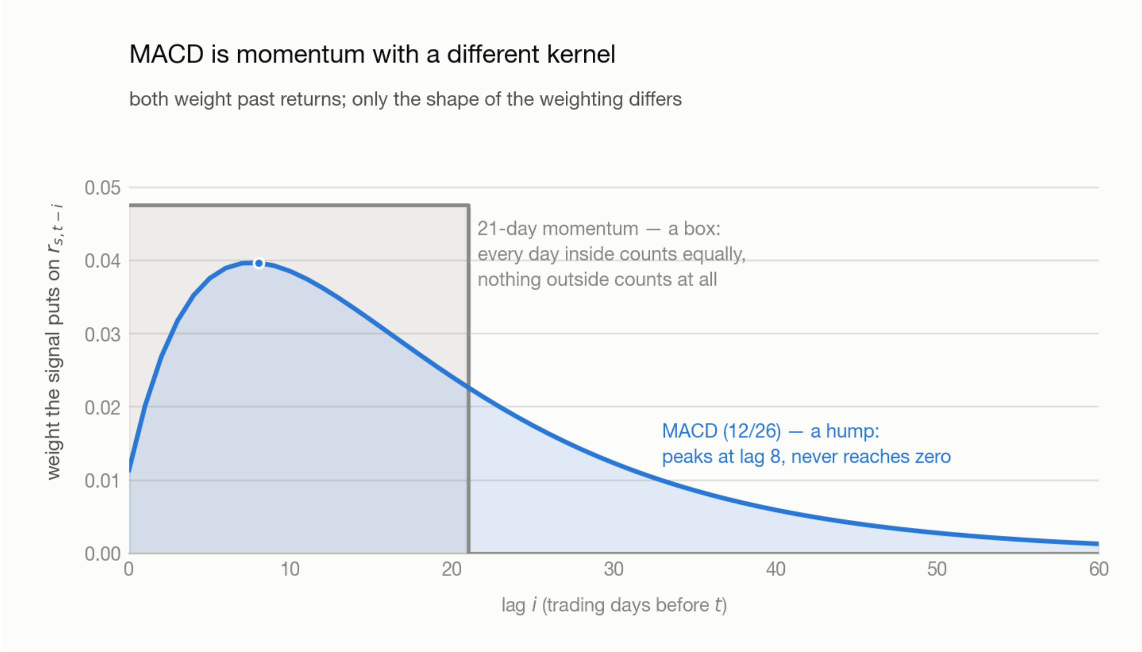

Figure 3.4: Weight given to each past daily return against its lag in trading days, for two signals normalised to the same total. A 21-day momentum is a flat box: equal weight for the last 21 days and zero beyond. MACD with spans 12 and 26 is a hump that starts low at lag zero, peaks around lag 8, then decays slowly and never reaches zero within sixty days. Illustrative.

Note (A choice, not a theorem)**.** §3.1.1 argued that the newest return deserves the most weight, and the most widely used trend indicator in the market does the opposite. Both positions are defensible: the newest return is the most recent evidence about the state the asset is in, and it is also the one carrying the most reversal (2's Background). Which wins is an empirical question settled by the bar plot, not a matter of taste—which is why §3.1.1 was written as a preference rather than a result.

3.3.3 From a value to a rule

Three common rules, in increasing order of information kept. All three still have to pass §2.3.2's bar plot before they earn a backtest.

Rule	**Go long when**	**Costs**
Zero-line	MACD is above zero	a slow trend filter, late
Crossover	the histogram is above zero	earlier, noisier—the churn is §3.4's subject
Proportional	always, sized by the standardized value	none of the magnitude, but see §2.4

The three differ in how much of the signal's magnitude they keep. They also differ in how *often* they trade, and that second axis is not a property of the rule alone: a crossover flips whenever the two legs touch, and how often they touch is set by how large the market's moves happen to be this month. Every parameter above is now chosen, and that count is still not under their control.

3.4 Volatility clustering breaks both

Definition 3.5 (Volatility clustering)**.** Large moves are followed by large moves and small by small, **regardless of sign**. Returns themselves are close to uncorrelated from one day to the next, but their *absolute* values are strongly and persistently autocorrelated. Direction is unpredictable; **amplitude is not**—a market has stretches, days to months, where everything is simply bigger, without the underlying trend having changed at all.

Neither piece survives it. A short EWMA half-life and MACD's fast leg are the same object under two names—a short average of recent returns—and a short average's swings scale with the amplitude around it. Enter a cluster and the sign starts flipping repeatedly, not because the trend turned but because the noise grew.

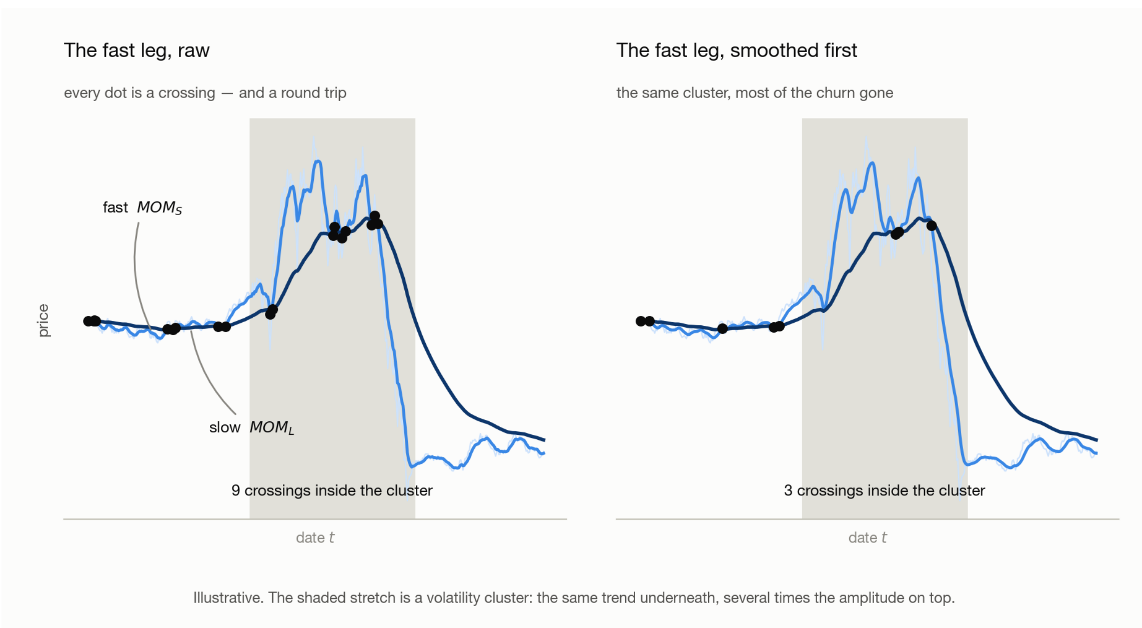

Figure 3.5: One price path, calm at both ends with a shaded volatility cluster in the middle, carrying a slow moving average and a fast one, every crossing marked. Left, the raw fast leg crosses the slow leg nine times inside the cluster. Right, the same fast leg passed through a short smoother first crosses three times. Illustrative.

The crossings are not merely expensive, they are backwards. A downward spike drags the fast leg under the slow one and flips the book short—at the bottom of the spike. The snap-back flips it long again, at the top. Repeat that through a cluster and the rule has been buying high and selling low on a schedule set by noise, which is worse than holding nothing: **whipsaw is not scatter around the right answer, it is the opposite of the right answer**, taken over and over. Nine crossings instead of three is nine round trips instead of three, and they land in exactly the stretch where spreads are widest and depth is thinnest—at forty crossings a year and five basis points a round trip, that is 200 bp of annual drag against a gross edge that might be four hundred.

The local fix is a shorter second average, or a margin. A **smoother**—a second average applied to the signal after it is computed, shorter than the signal's own period or it becomes another slow leg—stops a single day's noise from moving the sign on its own; MACD's 9-day signal line is one (§3.3.1). A **deadband**, which requires a crossing to clear some margin before the position flips rather than acting on the exact touch, is the other. Both read the signal and nothing else, so both are local: neither knows the tape has turned violent, each damps every date alike, and both pay on the quiet days to protect the loud ones. Going further means measuring the state of the market itself and asking what the rule is worth in each of its values.

→ 4 · Volatility Regimes.

Appendix · Notation

Throughout, t is the date. Everything in this chapter is computed one asset at a time, so the asset subscript s of 2 is dropped.

Symbol	**Means**	**First used**
a_t, λ	the newest observation, and the EWMA decay that keeps λ of the old average	§3.1.2
H	EWMA half-life, in periods	§3.1.3
N_f, N_s	fast and slow lookback lengths, in periods	§3.2.2
P_t, Δ_{t-j}	price on that date, and the one-period change at that lag	§3.3.1
n_f, n_s	fast and slow EMA spans, conventionally 12 and 26	§3.3.1
α_f, α_s	their smoothing constants, two over span plus one	§3.3.1
c_i, k_j	MACD's net weight on the price at that lag, and its kernel weight on the price change	§3.3.2

Note (Collisions to watch)**.** N_f, N_s are plain lookback lengths and n_f, n_s are the EMA spans that play the same two roles—case is the tell. k_j is a kernel weight, not a portfolio weight: 2 reserves $w_{s,t}$ for the latter, which is why the kernel is not written w. And α here is a smoothing constant, not §2.3.1's `alpha` plotting keyword and not a regression intercept; code font against maths is the tell.